6. It is mostly used in present day vehicles.
7. According to the up and down motion (18°) of propeller shaft the power is transmitted without any loss and vibrations.

4.8.2 Pot type universal joint

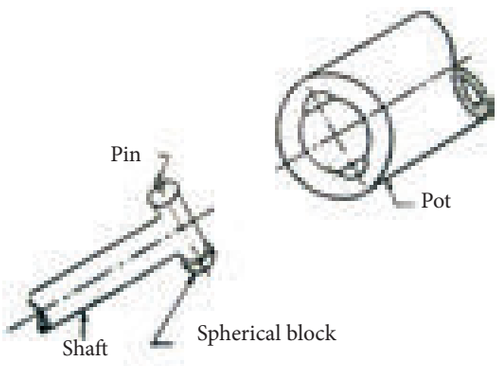


Figure 4.8.2 Pot type universal joint

One end of the shaft has T shape structure. Both the ends are joined with semi spheres using a pin. Cylinder like structure in the opposite is called pot type. Semi spheres in the T shaft are designed like moving part in the cylinder. So T shaft in the cylinder moves front and back and also in angles.

4.8.3 Rubber coupling type universal joint

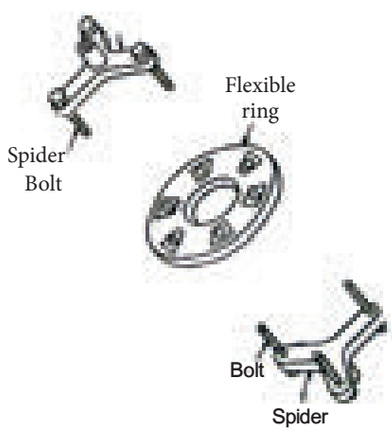
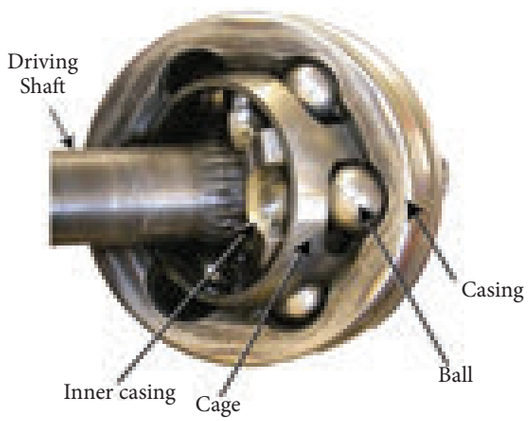


Figure 4.8.3 Rubber coupling type universal joint

In a round shaped rubber coupling there are six holes at an equal distance. This rubber coupling is placed between two spiders and fixed with bolt and nut. This rubber coupling is a flexible one to be with shaft. This can scope with only a small angle shake. It is also called as flexible ring type universal joint.

4.8.4 Ball type universal joint (Constant velocity type universal joint)



It is used to transmit the power approximately at 30° without vibration. Mostly used in front axle drive vehicles. It has input half shaft. One end of the shaft has splines in which inner spherical socket is fitted. Outer spherical socket is connected with the outer shaft. Grooves are cut between two sockets in which balls will move. When the input shaft rotates balls between the sockets will move and transmit the power smoothly.

4.9 Differential unit

When the vehicle moves straight on the road, the rear wheels run in the same way as the front wheels. The propeller shaft is connected to follow this. However, when the vehicle is negotiating a turn,

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